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DATA SCIENCE LAB FOR SMART CITIES

FINAL ESSAY

Hot and Poor: Mapping Heat Vulnerability Across Madrid's 131 Barrios

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Abstract

Extreme heat is one of the deadliest consequences of climate change in cities, and in Madrid its burden falls unevenly: the poorer and less vegetated barrios are measurably hotter. This project builds a reproducible Heat Vulnerability Index (HVI) for all 131 barrios of Madrid, combining seven years (2018–2024) of land surface temperature and vegetation data with official socioeconomic indicators. The resulting index reveals a structural pattern: the five most vulnerable barrios sit in Madrid’s historical working-class periphery, and the heat gap between the poorest and richest barrios has stayed positive in every single year of the study period. Spatial and multivariate analysis further shows that heat clusters geographically and that neighboring barrios exhibit significant spatial dependence in summer heat, that income and vegetation are among the strongest predictors of summer heat, and that not all “green” space actually cools: OpenStreetMap’s green-space tags are shown to be a poor proxy for real cooling compared with satellite-measured vegetation (NDVI). The HVI is positively correlated with the city’s official territorial vulnerability index ($r = +0.72$), providing an external consistency check. These findings translate into three concrete policy recommendations: target vegetation that genuinely cools, prioritize contiguous heat hotspots over isolated interventions, and pair every cooling measure with anti-displacement safeguards. Together, the results show that thermal vulnerability in Madrid is not randomly distributed, but layered on long-standing patterns of social and spatial inequality, and that effective climate adaptation must be designed with this in mind.

1 Problem Description and Indicators

1.1 Problem and societal significance

Extreme heat is gradually becoming one of the deadliest consequences of climate change in cities. This emergency is already a reality, not a distant projection. Between 2000 and 2019, approximately 489,000 deaths caused by heat occurred each year worldwide, with Europe responsible for about 36% of that total [1]. The summer of 2024 confirmed this trend at the continental level: a recent study estimated that heat contributed to approximately 62,775 deaths throughout Europe during the summer of 2024, a worrying heat emergency [2]. These figures show that extreme heat is more than a seasonal discomfort: it represents a major public health threat, particularly for vulnerable populations. This is not a danger confined to some future scenario; it is one we are already living through, summer after summer.

At the urban scale, Madrid provides a particularly relevant case. A comparative study by Arup (*Urban Heat Snapshot*, August 2023), which combined satellite imagery and AI-based climate modeling developed with the University of Reading, found that Madrid had the most severe urban heat island among the nine cities analyzed, with temperatures up to 8.5 °C higher than in surrounding rural areas [3]. Also, Sánchez-Guevara Sánchez et al. independently mapped Madrid's summer night-time urban heat island in detail and found a relationship between the spatial distribution of heat and the distribution of socially vulnerable populations across the city [4]. Together, these sources justify treating Madrid as a relevant case of urban heat inequality.

What this project aims to show is that urban heat is not just a physical or meteorological problem, but a public health issue whose risk is not evenly distributed throughout the city. Residents of poorer and less vegetated areas are more exposed to high temperatures than residents of other zones, and typically they also have fewer means of protecting themselves once that heat arrives: less access to air conditioning, or simply less ability to leave during the hottest periods. This link between physical exposure to heat and the social capacity to cope with it is the core of the project.

Previous studies have reported similar differences in heat exposure and vulnerability across Madrid and other Spanish cities. At the neighbourhood level, Sánchez-Guevara Sánchez et al. [4] found that Madrid's summer night-time urban heat island overlaps with the city's most socially vulnerable barrios, and a related study by the same Madrid research group, López-Bueno et al. [5], analysing heat-wave mortality through the city's districts, identified income level, the share of elderly residents and the availability of air conditioning among the key sociodemographic factors associated with mortality risk caused by heat. The health burden extends beyond mortality: Salvador et al. [6] showed that the risk of suffering a first cardiovascular event increased by 15.3% during extreme heat, with higher risk observed in more deprived neighbourhoods. Moreover, comparing urban and rural areas of the wider Madrid region, the same group [7], found greater vulnerability among the urban population than the surrounding rural one, with socioeconomic factors playing a central role in explaining the difference. A closely related issue is summer energy poverty: Sánchez-Guevara et al. [8] identified areas where high summer heat exposure overlaps with social vulnerability and housing risks, highlighting the vulnerability of poorer populations to summer energy poverty and cooling needs. Nor is the problem confined to Madrid borders: Marcińczak et al. [9] documented thermal inequalities over several Spanish cities, showing that the relationship between heat exposure and social characteristics extends beyond Madrid. Finally, even the most intuitive remedy carries its own risk: Anguelovski et al. [10] found that urban greening can be associated with processes of gentrification, potentially increasing displacement pressures and making the benefits of green improvements less accessible to socially vulnerable populations: a phenomenon commonly described as green gentrification.

Despite this body of research, the studies reviewed for this project do not provide a reproducible vulnerability index for all 131 Madrid barrios that combines heat exposure with official socioeconomic indicators at the barrio level. Addressing this gap is the contribution of the present project.

This raises four research questions that guide the rest of the analysis. How is heat distributed across Madrid's barrios? Is heat exposure correlated with social disadvantage and the absence of vegetation, producing a form of inequality? Once exposure, sensitivity and adaptive capacity are combined, which barrios turn out to be the most vulnerable? Where should adaptation ultimately be prioritized, so that it delivers equity, not merely efficiency?

1.2 Indicators and the design of the Heat Vulnerability Index

Following the framework set out by the IPCC [11], vulnerability is treated as a function of three dimensions: exposure, sensitivity, and adaptive capacity. Each dimension is measured through a small set of indicators, selected on the basis of data availability, reliability and relevance to the phenomenon under study.

1.2.1 Indicator selection and the composite index

- **Exposure** is measured as the average summer (June–August) land surface temperature (LST) over the period 2018–2024, derived from Landsat 8/9 Collection-2 imagery via Google Earth Engine.
- **Sensitivity** combines the share of the population aged 65+ residing in each barrio, population density and income level; the latter is weighted inversely, since a higher income is associated with lower sensitivity to heat.
- **Adaptive capacity** combines the Normalized Difference Vegetation Index (NDVI), used here as a proxy for the presence and density of vegetation relevant to cooling, with income, which reflects households' economic capacity to adapt to heat, for instance through air conditioning or home improvements.

Each indicator is normalised to the $[0, 1]$ range via min–max scaling. The composite index is then defined as

$$HVI = \frac{E + S + (1 - AC)}{3},$$

where E , S and AC denote the normalised exposure, sensitivity and adaptive capacity components respectively. Since each component lies in $[0, 1]$, so does the resulting HVI , with higher values indicating higher vulnerability.

1.2.2 Why NDVI, not OSM green space

An important methodological choice concerns how vegetation is measured within the adaptive capacity component. A correlation analysis between two candidate measures and LST reveals a divergence. The share of land classified as green space by OpenStreetMap correlates weakly and positively with LST ($r = +0.13$ Pearson, $+0.15$ Spearman), whereas NDVI correlates negatively and more strongly ($r = -0.52$ Pearson, -0.52 Spearman), consistent with the expected cooling power of vegetation. This may be because OpenStreetMap classifies large peripheral areas that are dry or uncultivated as green land, even though they provide limited cooling and are among the hottest parts of the city, while NDVI captures the presence and intensity of active vegetation, making it a better proxy for vegetated surfaces relevant to cooling. The cooling benefits of urban green infrastructure depend on its quantity but also on its cooling efficiency [12].

1.3 Ethical and social implications

1. **Risk of territorial stigmatisation.** Publishing an index that explicitly labels certain barrios as “more vulnerable” carries the risk of reinforcing an existing territorial stigma, particularly in areas that are already socially marginalised, such as Casco Histórico de Vallecas, which is the barrio ranked as most vulnerable in the results of this project. For this reason, the index must be framed and used strictly as a tool for directing resources and targeting adaptation efforts towards residents, never as a label to penalise a community or depress local property values.
2. **Data limitations and the normativity of the index.** Land surface temperature is derived from a daytime satellite image and measures the temperature of the ground surface, not the air temperature that people actually experience, also it is affected by cloud cover and by the exact time of the satellite overpass; it should be treated as a proxy for heat exposure rather than a direct measurement of thermal experience. Moreover, giving equal weight to the three dimensions of the composite index is a normative choice. A PCA-weighted version of the index is therefore included as a robustness check. It is used alongside the primary HVI not as a replacement.
3. **Risk of maladaptation: green gentrification.** A further risk is green gentrification, as documented [10]. Increasing vegetation is an obvious adaptation measure, but urban greening can also be associated with green gentrification: new green amenities may increase the attractiveness and property values of an

area, potentially contributing to higher rents and displacement pressures on poorer residents. This means that equity must be designed into an intervention from the outset. Green-based adaptation measures targeted at Madrid's most vulnerable barrios should be accompanied by anti-displacement safeguards, rather than being considered just in terms of their potential to increase urban vegetation.

2 Data Analytics, Optimization and Policy Suggestions

2.1 Data and data wrangling

Four types of data were combined: administrative boundaries, socioeconomic indicators, green-space indicators, and land surface temperature together with a vegetation index.

The first dataset consists of Madrid’s official administrative boundaries, obtained from the Ayuntamiento de Madrid’s open data Geoportal [13]. The city of Madrid is divided into 21 *distritos* (districts), its main administrative subdivisions, each of which is in turn divided into *barrios* (neighbourhoods), 131 in total. The original column names were renamed to English for consistency: `district_code` and `district_name` identify the district a barrio belongs to, `barrio_code` and `barrio_name` identify the barrio itself, and `district_barrio_code` is the combined identifier, used as the join key to merge every other dataset onto this same spatial unit. The dataset also carries, for every barrio and district a polygon geometry describing its exact administrative boundary. As a first data quality check, the boundaries were confirmed to cover all 131 barrios grouped into 21 districts, with a total area of about 604 km², matching the known surface area of the municipality of Madrid. Geometry validity was verified for all 131 polygons; none required repair in this case. All geometries were then reprojected to EPSG:25830 (ETRS89 / UTM zone 30N), a metric coordinate reference system needed to compute area, and later population density, correctly.

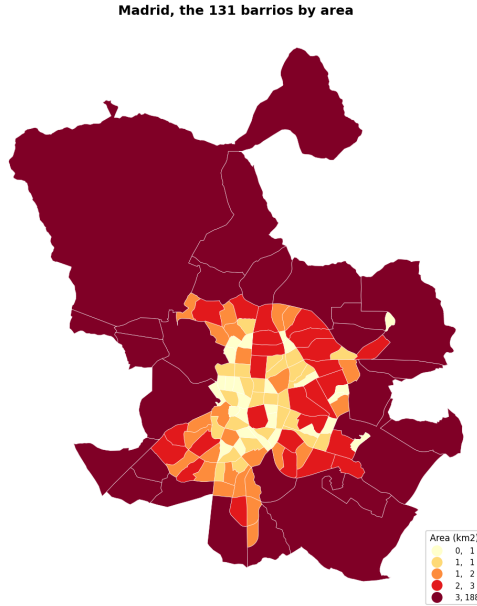


Figure 1: Madrid’s 131 barrios, coloured by area (km²).

The second dataset consists of socioeconomic indicators, obtained from the Ayuntamiento de Madrid’s own open-data panel of district and barrio indicators [14]. The seven barrio-level variables used are: `population`, the number of residents; `pct_65plus`, the share of residents aged 65+; `income_per_person` and `household_income`, the average disposable income per person and the average net annual household income respectively, and three official composite vulnerability indices published by the city: `vuln_health` and `vuln_environment`, covering the health and the urban-environment and mobility spheres of the municipality’s own vulnerability framework, and `vuln_territorial`, a broader aggregated territorial vulnerability score which the city computes by combining social welfare and equality, urban environment and mobility, education and culture, economy and employment, and health into a single weighted index.

Each indicator in the panel is updated with its own schedule, so different variables have different reference years. Rather than restricting the dataset to the oldest year common to all seven indicators, which would have discarded more recent, the most recent year available was kept for each indicator individually, maximising the completeness and freshness of the data at the cost of a minor inconsistency in reference years across variables.

The resulting indicator table was then merged onto the barrio boundaries via a name-based join, matching

on normalised barrio names; because the boundaries layer is a GeoDataFrame, this merge preserves each barrio's polygon geometry. No barrio was left unmatched.

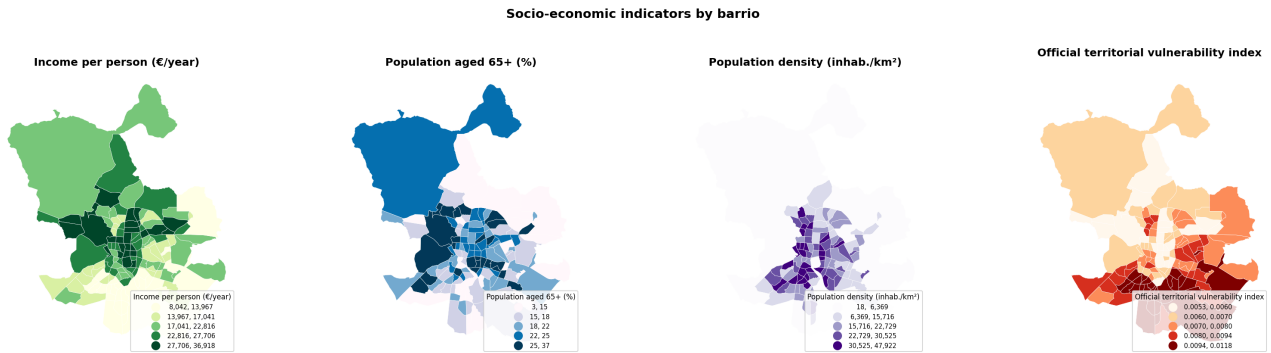


Figure 2: socioeconomic indicators by barrio: income per person, share of residents aged 65+, population density, and the official territorial vulnerability index.

The third dataset covers urban green space, obtained live from OpenStreetMap, querying all features falling within Madrid's boundary. Green space was defined using a fixed list of OpenStreetMap tags: **leisure** = park, garden, nature reserve, recreation ground or dog park; **landuse** = grass, forest, meadow, recreation ground, village green, cemetery or greenfield; and **natural** = wood, grassland, scrub or heat. The live query returned 20,128 raw features; any invalid or self-intersecting polygons were repaired with the standard `buffer(0)` trick, and overlapping green polygons were dissolved into a single shape to avoid double counting a park that happens to be split across multiple elements. The resulting green layer was then intersected with each barrio's boundary. Three metrics were constructed: **green_area_m2**, the green area falling inside each barrio; **pct_green**, the share of the barrio's total area that is green; and **green_m2_per_cap**, green area per resident.

A known limitation of this OpenStreetMap's measure is what can be called *dry green*: some of the tags used to define green space can also apply to large, dry, uncultivated or rocky areas at the city's edge, which are counted as green space here even though they may not actually provide much cooling. This is a limitation of the underlying classification and is worth keeping in mind when interpreting **pct_green** and **green_m2_per_cap** in what follows.

Green space is also distributed very unevenly across the city. **pct_green** ranges from just 0.7% in Goya and 0.8% in Sol (both dense, central barrios) up to 96.5% in El Pardo and 85.9% in Casa de Campo, two barrios dominated by Madrid's largest parks and forests. Green space per resident is even more skewed: the city mean (473 m² per person) is driven almost entirely by a handful of barrios adjacent to parks and not very populated, while the median barrio has only about 8 m² of green space per resident.

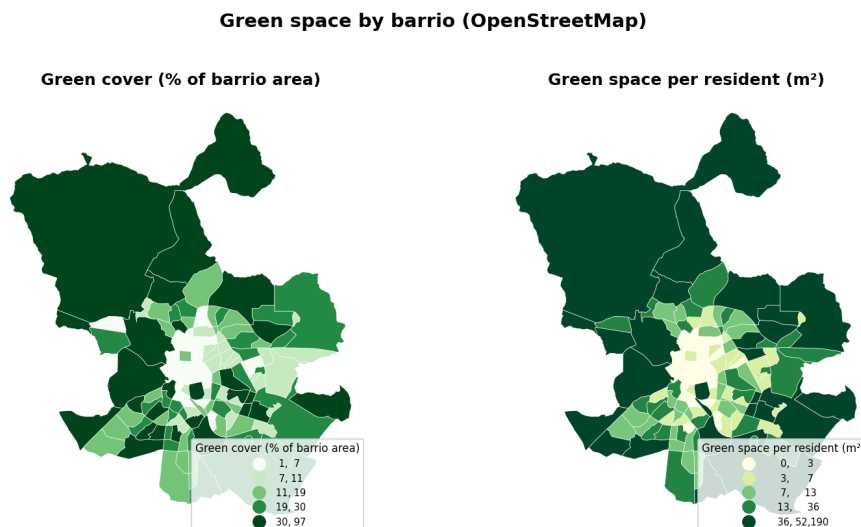


Figure 3: Green-space cover (**pct_green**) and green space per resident (**green_m2_per_cap**) by barrio.

The fourth dataset measures heat exposure directly from satellite imagery: Landsat 8 and 9, Collection-2 Level-2 data (USGS/NASA), accessed through Google Earth Engine.

Two indices are derived each as a per-pixel median composite over the full 2018–2024 period rather than from a single year, which gives a more robust estimate of a typical summer and avoids the result being skewed by any one exceptionally hot or cool year:

- **LST_sum** (Land Surface Temperature): the median summer (June–August) land surface temperature over 2018–2024, in degrees Celsius. Across the 131 barrios it ranges from 37.05 to 48.86 °C (mean 42.85 °C, std 1.73).
- **NDVI_sum** (Normalized Difference Vegetation Index): the median summer vegetation index over the same period, ranging in principle from -1 to 1 , with higher values indicating denser, healthier vegetation. Across the 131 barrios it ranges from 0.12 to 0.49 (mean 0.26, std 0.07).

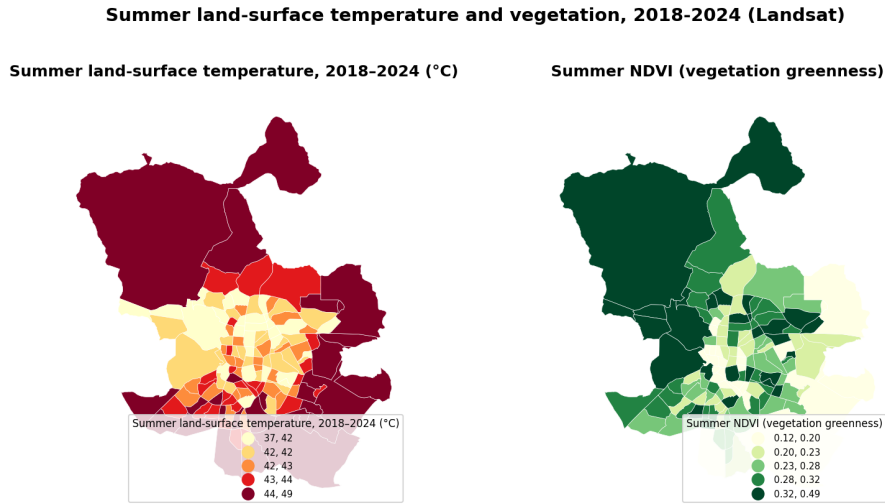


Figure 4: Land surface temperature (LST_sum) and vegetation greenness (NDVI_sum) by barrio, 2018–2024.

The final step of the data preparation consists of merging the three layers just constructed (socioeconomic indicators, green space, and land surface temperature/NDVI) onto the barrio boundaries, all joined on `district_barrio_code`. The result is a single master table with 131 rows, one per barrio, 27 columns, and no missing values.

A first check of this master table is to look directly at which barrios are hottest and which are coolest, summarised in Table 1 and Table 2.

Table 1: The eight hottest barrios in Madrid by median land surface temperature (2018–2024), with income, vegetation (NDVI) and green space share.

Rank	Barrio	District	LST (°C)	Income (€)	NDVI	% green
1	Casco Histórico de Vallecas	Villa de Vallecas	48.9	11,951	0.19	23.5
2	Campamento	Latina	47.4	14,656	0.30	63.5
3	Aeropuerto	Barajas	47.4	12,420	0.20	25.9
4	El Cañaveral	Vicálvaro	47.0	19,187	0.20	40.4
5	Cuatro Vientos	Latina	46.9	17,516	0.24	16.6
6	Villaverde Alto: Casco Histórico de Villaverde	Villaverde	46.7	11,511	0.18	18.1
7	Butarque	Villaverde	46.5	13,620	0.21	24.0
8	Casco Histórico de Vicálvaro	Vicálvaro	46.5	12,291	0.18	25.4

Table 2: The eight coolest barrios in Madrid by median land surface temperature (2018–2024), with income, vegetation (NDVI) and green space share.

Rank	Barrio	District	LST (°C)	Income (€)	NDVI	% green
1	Los Jerónimos	Retiro	37.0	31,815	0.49	69.4
2	Atalaya	Ciudad Lineal	39.8	33,140	0.39	11.3
3	Nueva España	Chamartín	40.2	34,371	0.32	9.9
4	San Juan Bautista	Ciudad Lineal	40.2	28,260	0.33	9.0
5	Costillares	Ciudad Lineal	40.2	29,377	0.34	14.7
6	Palacio	Centro	40.2	23,770	0.33	34.7
7	Legazpi	Arganzuela	40.4	25,356	0.35	38.7
8	Hispanoamérica	Chamartín	40.5	31,917	0.30	4.8

Campamento provides a good example: it is the second hottest barrio in Madrid, but also has an unusually high share of green cover (63.5%). It illustrates the dry green problem introduced earlier: it contains large tracts of dry land that OpenStreetMap classifies as green space, even though this vegetation provides almost no real cooling.

2.2 Spatio-temporal and correlation analysis

The next thing to show is how summer heat is distributed across Madrid’s barrios and test it against the hypotheses to establish whether that exposure amounts to a measurable form of inequality: whether hotter barrios are also poorer, whether green space lowers heat, despite being unevenly distributed, and whether population density and the age structure of a barrio’s population play a meaningful role.

Figure 5 reports the Pearson correlation between summer land surface temperature (`LST_sum`) and each candidate driver, together with the correlations among the drivers themselves, across all 131 barrios.

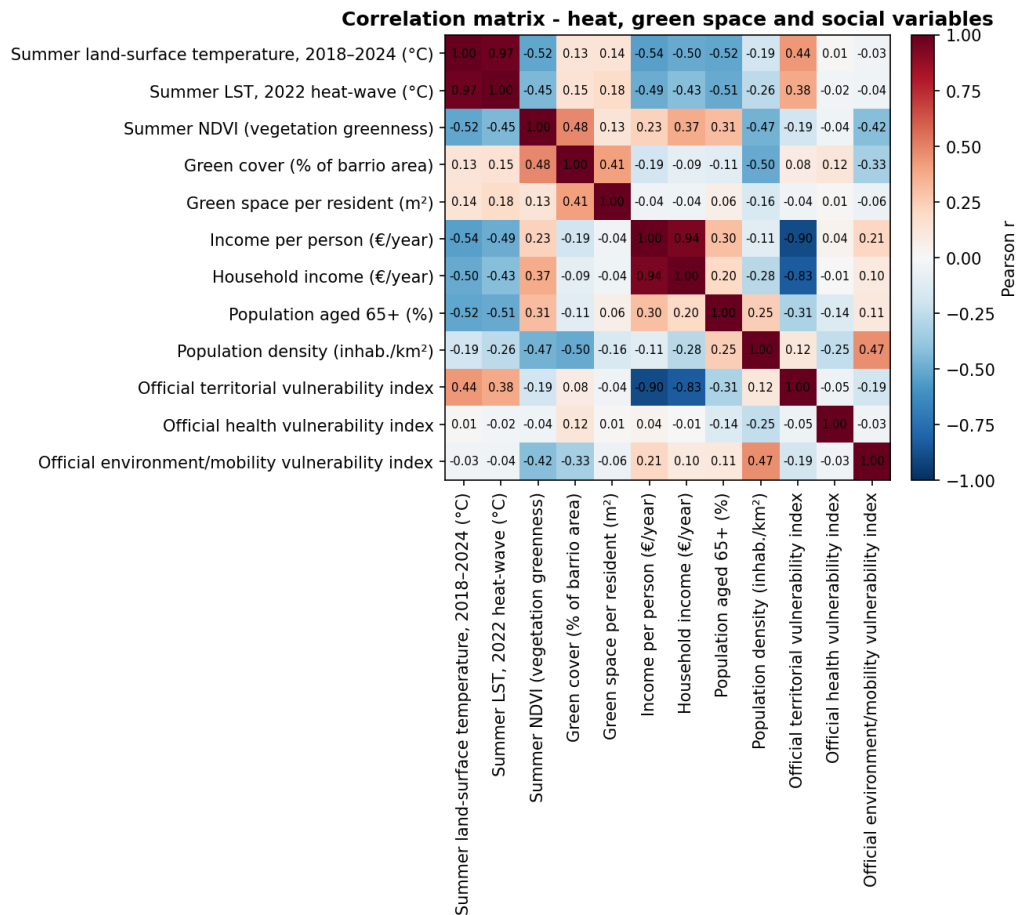


Figure 5: Correlation matrix of land surface temperature, green space and socioeconomic variables (Pearson r , 131 barrios).

The pattern is consistent. Income, NDVI and the share of residents aged 65+ all correlate negatively with heat: barrios that are richer, greener and have an older population tend to be cooler. The city's own official territorial vulnerability index, by contrast, correlates positively with heat, reinforcing the link between social disadvantage and thermal exposure that motivates this project. Green space share, however, tells a very different story: it correlates only weakly, and counter-intuitively positively, with heat. This confirms the dry green problem already flagged during data preparation: the issue lies in how the underlying data classifies land, not in interpretation, and it is why NDVI, rather than green cover, is used as the vegetation measure.

These same relationships are visible in the data, barrio by barrio, in Figure 6, which plots LST against its four main candidate drivers together with a linear trend for each.

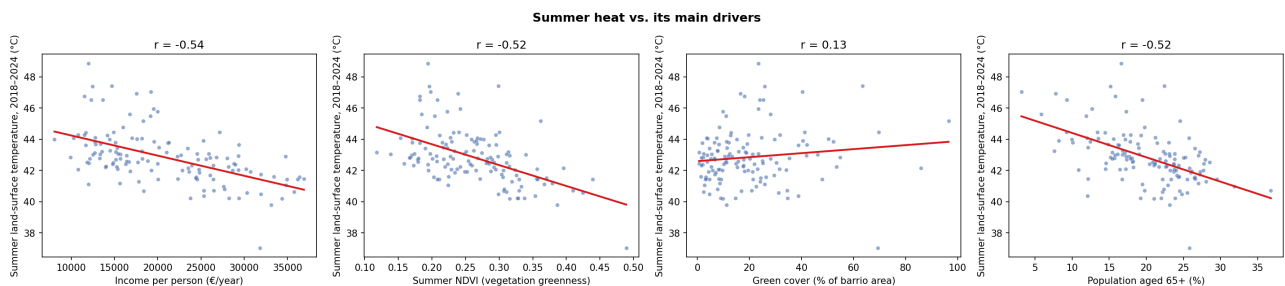


Figure 6: Land surface temperature against income per person, NDVI, green space share, and the share of residents aged 65+, with a linear trend fitted to each.

Correlation alone cannot say whether hot barrios are clustered together in space or simply scattered at

random; answering that requires methods built specifically to identify spatial patterns treating the spatial relationships between barrios as data in their own right. For this reason, the analysis is extended with two spatial-autocorrelation statistics, both computed on a Queen contiguity spatial-weights matrix, in which two barrios are treated as neighbours if they share a boundary or even just a corner point.

Global Moran’s I [15] tests whether hot barrios sit disproportionately next to other hot barrios across the whole city: a positive and statistically significant value means heat is spatially clustered rather than randomly distributed. For LST, Moran’s $I = 0.418$ ($p = 0.001$), a statistically significant signal that heat clusters spatially: hot barrios tend to sit next to other hot barrios.

Local Indicators of Spatial Association (LISA) [16] go a step further, classifying each individual barrio according to how its own heat level relates to that of its neighbours: a High-High barrio is hot and surrounded by hot neighbours; a Low-Low barrio is cool and surrounded by cool neighbours; High-Low and Low-High barrios are local outliers that stand out against the surrounding area; and barrios with no statistically significant relationship to their neighbours are labelled not significant. Applied to LST, LISA identifies 18 High-High barrios, 25 Low-Low barrios, 4 High-Low outliers and 2 Low-High outliers, with the remaining 82 barrios showing no significant local pattern.

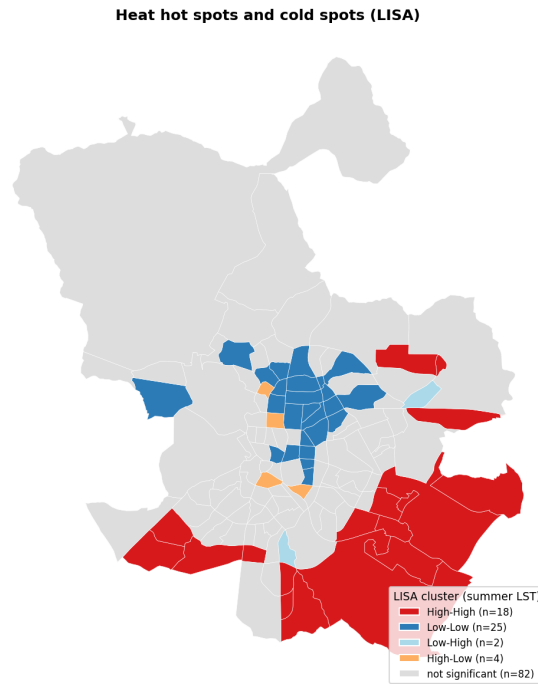


Figure 7: LISA cluster map of land surface temperature: heat hot spots (High-High) and cold spots (Low-Low) across Madrid’s barrios.

The yearly data can be used to assess whether the heat gap between richer and poorer barrios persists over time. Figure 8 addresses this, tracking summer LST through all seven study years (2018–2024) for the city as a whole, the poorest income quartile of barrios, and the richest income quartile.

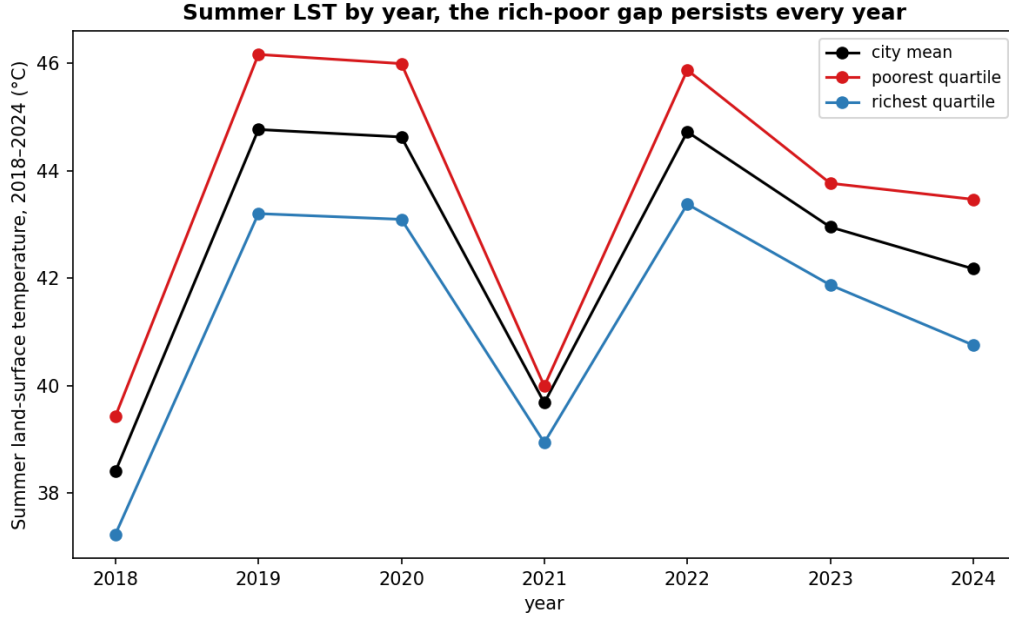


Figure 8: Land surface temperature by year, city-wide and split by the poorest and richest income quartiles of barrios (2018–2024).

The gap between the poorest and richest quartile is positive in every single year of the study period, averaging $+2.32^{\circ}\text{C}$, with the narrowest gap recorded in 2021 ($+1.05^{\circ}\text{C}$). The size of the gap fluctuates from year to year, reflecting differences in how hot or cool each individual summer was, but its sign never reverses: the poorest barrios are consistently hotter than the richest one. This is empirical evidence that the thermal inequality observed across Madrid’s barrios is structural, not just a coincidence tied to any single summer.

2.3 Modelling

This section puts into practice the Heat Vulnerability Index (HVI): a composite measure of exposure, sensitivity and adaptive capacity,

$$HVI = \frac{E + S + (1 - AC)}{3},$$

where E is built from summer land surface temperature; S from the share of residents aged 65+, population density, and income (inversely, since lower income increases sensitivity); and AC from NDVI and income, each rescaled to $[0, 1]$ before being combined, so that a higher HVI always denotes greater vulnerability. Table 3 reports the resulting index, and its three underlying components, computed for all 131 barrios of Madrid.

Table 3: Descriptive statistics of the Heat Vulnerability Index and its three components, across all 131 barrios (all values rescaled to $[0, 1]$).

Component	Mean	Std	Min	25%	Median	75%	Max
Exposure (E)	0.491	0.147	0.000	0.409	0.481	0.566	1.000
Sensitivity (S)	0.488	0.140	0.105	0.389	0.529	0.591	0.729
Adaptive capacity (AC)	0.415	0.171	0.086	0.290	0.380	0.502	0.920
HVI	0.627	0.195	0.000	0.556	0.672	0.749	1.000

The mean HVI across Madrid is 0.627, and its median (0.672) is higher than its mean, indicating a distribution skewed towards higher vulnerability: more barrios sit on the vulnerable half of the scale than on the safe half. Figure 9 maps the index.

Heat Vulnerability Index - Madrid barrios

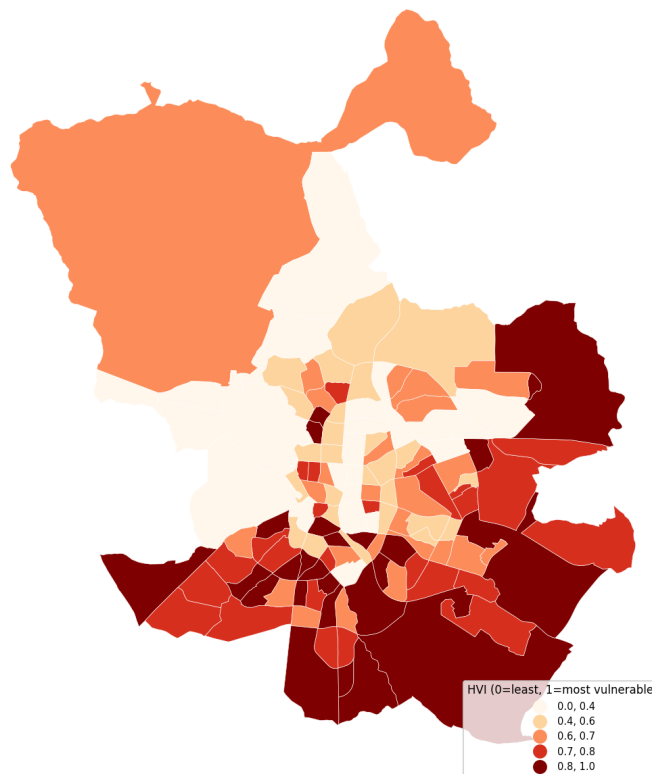


Figure 9: Heat Vulnerability Index across Madrid's 131 barrios (0 = least vulnerable, 1 = most vulnerable).

The five most vulnerable barrios in Madrid, by HVI, are:

1. **Casco Histórico de Vallecas** (Villa de Vallecas): HVI = 1.000, the single most vulnerable barrio in the city, combining the highest LST recorded anywhere in Madrid (48.9 °C) with one of the lowest incomes per person (€11,951).
2. **San Diego** (Puente de Vallecas): HVI = 0.989, with the lowest income per person of any barrio in Madrid (€10,252) and a LST of 44.1 °C.
3. **Villaverde Alto: Casco Histórico de Villaverde** (Villaverde): HVI = 0.932, LST of 46.7 °C and income of €11,511 per person.
4. **Casco Histórico de Vicálvaro** (Vicálvaro): HVI = 0.914, LST of 46.5 °C and income of €12,291 per person.
5. **Aeropuerto** (Barajas): HVI = 0.913, the barrio bordering Madrid-Barajas Airport, with a LST of 47.4 °C, the highest of the five, and income of €12,420 per person.

The ranking also reveals that: three of the five most vulnerable barrios in Madrid, Casco Histórico de Vallecas, Villaverde Alto–Casco Histórico de Villaverde, and Casco Histórico de Vicálvaro, carry *Casco Histórico* (“historic core”) in their name. This isn’t a coincidence, but a trace of Madrid’s own urban history [17]. Vallecas, Villaverde and Vicálvaro were independent municipalities, annexed to the city of Madrid in the mid XX century. Before annexation, these were predominantly rural or industrial, working-class settlements. Once absorbed into the city, each former village’s old centre was preserved administratively as a distinct barrio and given the *Casco Histórico* label, while their historical development contributed to the social and spatial patterns that still distinguish parts of Madrid’s southern and south-eastern periphery. The fact that the same barrios resurface today at the top of an index built independently from satellite heat data and open socioeconomic indicators, with no reference to this administrative history, suggests that today’s thermal vulnerability in Madrid is layered on top of long standing patterns of urban and social marginalisation.

As a robustness check, the same standardised indicators (LST, share aged 65+, population density, NDVI, income) are combined instead through Principal Component Analysis (PCA): their first principal component, sign-aligned so that it increases with heat, defines a data-driven version of the index, HVI_{PCA} , with no normative weighting choice imposed. The two versions of the index agree strongly ($r = 0.833$ across all 131 barrios), indicating that the barrio ranking obtained reflects a pattern that is also present in the underlying data.

2.3.1 Explanatory model: what drives summer heat?

The analysis also examines which variables are most strongly associated with summer heat.

The same five variables that underlie the HVI, income, population density, share aged 65+, NDVI, and the green space share, are used as predictors: in the OLS regression below, in the spatial lag model that follows it, and in the Random Forest model further on.

The relationship is estimated using an OLS regression of LST on these five predictors. All predictors are standardised, so the coefficients show the change in LST, in degrees Celsius, for a one standard deviation change in each predictor. This makes predictors measured in different units, such as euros, percentages, and inhabitants per km^2 , directly comparable. Table 4 reports the results.

Table 4: OLS regression of LST on standardised predictors ($R^2 = 0.722$, adjusted $R^2 = 0.711$, $N = 131$). Coefficients denote the change in LST, in $^{\circ}C$, per one standard deviation change in each predictor.

Predictor	Coefficient ($^{\circ}C$)	p -value
NDVI	-1.247	< 0.001
Population density	-0.844	< 0.001
Income per person	-0.668	< 0.001
OSM green-space share	+0.264	0.016
Share aged 65+	-0.086	0.398 (not significant)

NDVI emerges as the strongest driver of summer heat, ahead of population density and income. The green space share is again positively, even if only weakly, associated with heat ($p = 0.016$): the same dry green pattern confirmed within a multivariate model. The share of residents aged 65+ is not statistically significant ($p = 0.398$) once income, density and vegetation are accounted for.

A standard OLS regression assumes each barrio’s heat is independent of its neighbours’, an assumption contradicted by the significant spatial clustering already shown. A spatial lag model addresses this by adding the mean LST of each barrio’s neighbours as an additional predictor, using the same Queen contiguity weights matrix as the Moran’s I/LISA analysis, so that the remaining coefficients are not inflated by this spatial spillover.

Because of this spatial feedback loop, each barrio’s heat depends on its neighbours’, whose heat in turn depends on their own neighbours’, and so on, a change in a single barrio’s own characteristics does not only affect that barrio’s own heat: it also propagates outward, influencing its neighbours and, more weakly, its neighbours’ neighbours. The spatial lag model decomposes this total impact into three separate quantities for each predictor: the *direct effect*, the impact of a change in a barrio’s own predictor on its own LST; the *indirect effect*, the cumulative impact of that same change on every other barrio in the city, transmitted through the spatial network of neighbours; and the *total effect*, the sum of the two. Table 5 reports all three for each predictor.

Table 5: Direct, indirect and total effects of each predictor on LST, from the spatial lag model (units: $^{\circ}C$ per one standard deviation change in the predictor).

Predictor	Direct	Indirect	Total
NDVI	-1.244	-0.619	-1.864
Population density	-0.714	-0.355	-1.069
Income per person	-0.478	-0.238	-0.716
OSM green-space share	+0.262	+0.131	+0.393
Share aged 65+	+0.023	+0.011	+0.034

The indirect effect is consistently about half the size of the direct effect for every predictor, confirming that spillovers across barrio boundaries are a non negligible part of how these variables shape heat. NDVI, for instance, cools a barrio by 1.244 °C per standard deviation, but also cools its surrounding barrios by a further 0.619 °C in total, for a combined city wide effect of 1.864 °C.

2.3.2 Predictive model: Random Forest

The models above are *explanatory*: each coefficient says how many degrees LST changes for a one standard deviation change in that predictor, holding the others fixed. A Random Forest instead builds a *predictive* model: it grows many decision trees that repeatedly split barrios by whichever variable best separates them by LST, and reports how much each variable contributes to those splits, its *feature importance*. It says how useful a variable is for predicting LST accurately. To evaluate the model fairly, its R^2 is computed by 5-fold cross-validation: the model is tested on barrios it was not trained on.

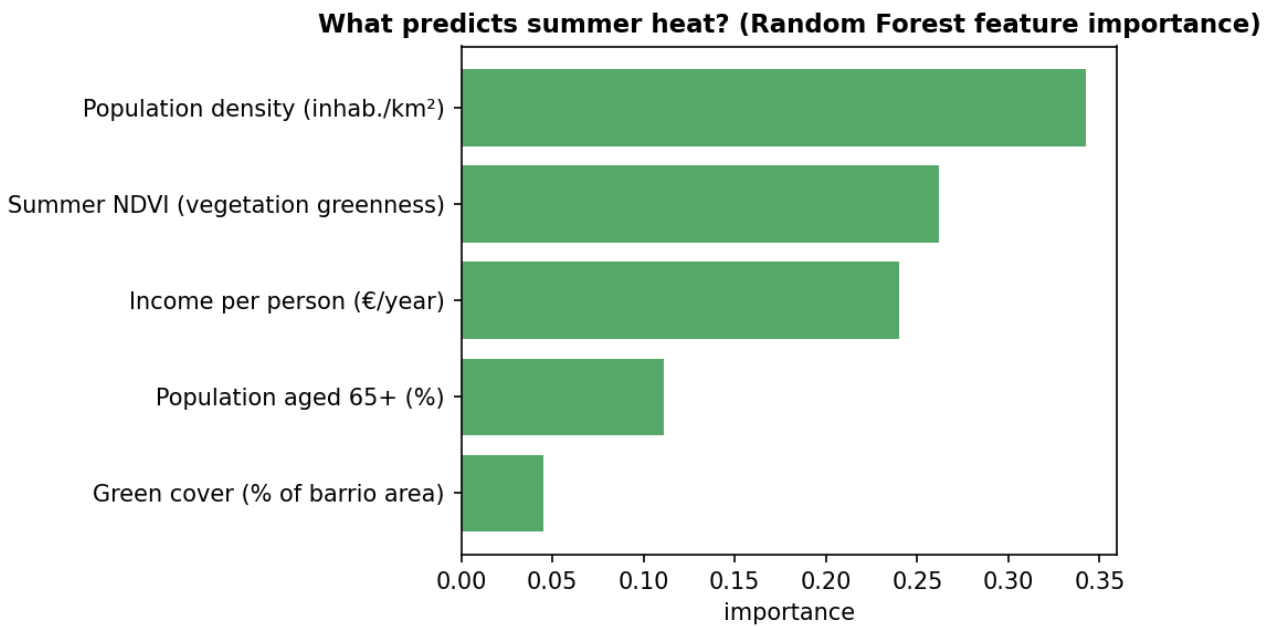


Figure 10: Random Forest feature importance for predicting land surface temperature.

Comparing the Random Forest results with the OLS coefficients highlights several differences between the two approaches. Population density is the single most important predictor for the Random Forest, ahead of NDVI; and the share of residents aged 65+ ranks fourth out of five in importance, ahead of the green space share. The difference reflects the fact that the two models capture different types of relationships: a Random Forest can pick up non-linear relationships and interactions between variables (for instance, age structure mattering more in combination with density or income than on its own) that a linear model cannot represent as a single coefficient. Together, the two models point to income, population density and vegetation as the main predictors associated with heat in Madrid.

2.3.3 Clustering: barrio typologies

A complementary question is whether Madrid’s barrios fall into a small number of recognisable and interpretable types. This is addressed with k -means clustering, with $k = 4$: clustering has no dependent variable to keep separate, since its purpose is to group barrios by overall similarity rather than to predict or explain one variable from the others. Table 6 reports the resulting typologies.

Table 6: Mean characteristics of the four barrio typologies found by k -means clustering ($k = 4$). For each variable, the highest value across the four clusters is shown in bold and the lowest is underlined.

Cluster	n	LST ($^{\circ}\text{C}$)	NDVI	Income (€)	% 65+	Density (inhab./ km^2)	% green	HVI
0	31	42.57	<u>0.19</u>	25,019	21.4	33,086	<u>5.0</u>	0.66
1	23	45.48	0.23	16,660	<u>13.3</u>	<u>4,918</u>	32.6	0.74
2	27	<u>41.06</u>	0.35	29,326	22.5	9,800	24.9	<u>0.33</u>
3	50	42.77	0.27	<u>15,452</u>	20.8	23,032	20.4	0.72

The four clusters can be characterised as follows:

- **Cluster 0 (dense, low green, middle income; 31 barrios)** is Madrid’s dense core: by far the highest population density of the four, and, both the lowest NDVI *and* the lowest green share, with a relatively high income;
- **Cluster 1 (hot, poor, dry green periphery; 23 barrios)** is the hottest typology, combining the lowest income and population density with the highest green share. It also has the highest HVI mean, making it the most vulnerable typology;
- **Cluster 2 (cool, green, rich; 27 barrios)** is the coolest, richest and greenest typology, and by a wide margin the least vulnerable;
- **Cluster 3 (dense, low income, moderate heat; 50 barrios)** is the largest typology, accounting for more than a third of all barrios, combining the lowest income of the four with fairly high density and only moderate heat. It is the second most vulnerable typology.

Barrio typologies (k-means, 4 groups)

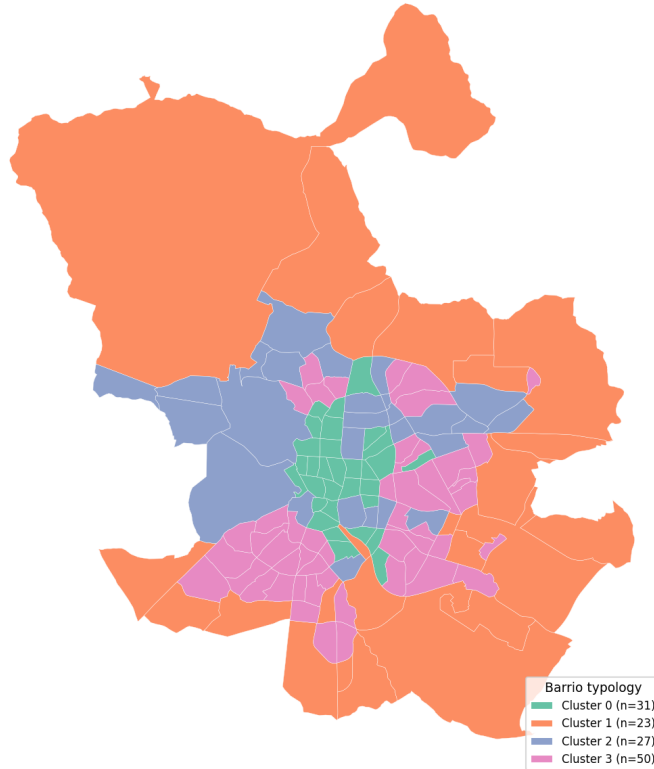


Figure 11: Barrio typologies found by k -means clustering ($k = 4$).

A bivariate map makes the resulting inequality visible in a single view. Figure 12 classifies each barrio jointly by its heat and its poverty tercile, so that the darkest colour marks barrios that are simultaneously hot *and* poor.

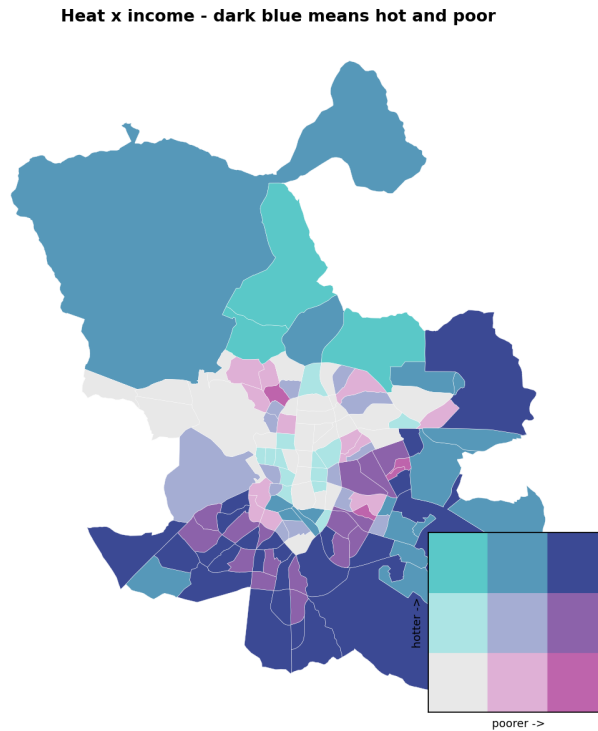


Figure 12: Bivariate map of heat and poverty by barrio tercile; the darkest colour marks barrios that are both hot and poor.

2.3.4 Validating the HVI

A final check is whether the HVI agrees with evidence external to this project. The HVI correlates at $r = +0.72$ with the city's own official territorial vulnerability index, at $r = -0.84$ with income per person, and at $r = +0.83$ with its own PCA-weighted variant. The agreement with the official municipal index provides an external consistency check: the two indices are constructed from different indicators and methodologies, but their correlation suggests that the HVI reflects the municipality's own findings.

2.4 Policy recommendations

The analysis provides evidence on how heat is distributed across Madrid and on where adaptation could be prioritised to improve equity. The analysis assembled above points to where, how and for whom heat adaptation in Madrid should be designed. Three recommendations follow from the evidence.

1. **Prioritise vegetation with measurable cooling effects.** NDVI has consistently outperformed the green space share as a predictor of cooling, and the dry green problem it exposes has been confirmed at every level of this analysis. This points to a policy implication, echoing Li et al. [12]: greening programmes should be designed around the real cooling potential (dense tree canopy and living vegetation with measurable NDVI gains). Campamento illustrates this well: planting or preserving dry scrubland or drought tolerant ground and, counting it as a climate intervention, risks spending public resources on measures that barely lower actual land surface temperature.
2. **Prioritise connected areas of heat rather than individual barrios.** The spatial lag model found that heat conditions in a given barrio are associated with those of the neighbouring's one. The practical implication is to consider these contiguous hotspot clusters as connected zones when prioritising adaptation investments, rather than treating each barrio as an isolated unit. Multiple small investments across

adjacent barrios could generate a greater collective cooling effect due to the heat spills over between neighbours.

3. **Pair every intervention with anti-displacement safeguards.** The barrios with the highest HVI are systematically also among the poorest: Casco Histórico de Vallecas, the most vulnerable barrio in the city, has one of the lowest incomes per person. Cooling an already vulnerable barrio without safeguards risks to hurt the people that it was supposed to protect: new vegetation and public space can raise an area's attractiveness and property values, and this could price out the poorest residents (green gentrification). Every cooling intervention should be paired with concrete anti-displacement guarantees, for example rent protections and genuine community involvement in project design, rather than treating equity as an afterthought to be addressed only if problems emerge later.

Overall, adaptation should focus on areas where cooling measures can be effective, where neighbouring barrios may also benefit, and where interventions can be implemented without increasing displacement pressures.

Code and data availability. All code used in this project is available at:

https://drive.google.com/drive/folders/19IdCrR4dZjfC33CVYXLbLYtIp2Su8BJn?usp=share_link

Declaration of Generative AI Use

The use of Generative AI in this project is disclosed as follows. All ideas, analytical choices, interpretation of results, and the writing of the essay's content are the author's own. Generative AI was used strictly as a support tool, in the following ways: (i) as a research aid to identify and locate relevant scientific literature discussed in this report, and to help format the corresponding bibliography entries in the required citation style; (ii) for grammar and language refinement of the English text; (iii) to assist in formatting and laying out the tables and figures within the LaTeX document; and (iv) as a support tool in the development of the Python code underlying Part 2 of the project.

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